Practical based on A5-R5.1, A6-R5.1 and A7-R5.1

Max. Marks 100 = 80(Practical) + 20(Viva Voce)

Duration of Practical Examination 3Hrs duration including viva-voce

A5-R5.1 (Data Structure)

(ii) Write a C++ program that reads a number from the user and counts the number of digits and sum of digits of the number

(iii) Create a class in C++ called rational1 that stores numerator and denominator. Write the functions to input the data (ensure that denominator is not equal to 0), to display the number. Also use operator overloading to overload the arithmetic operators: +,-,*,/ and also relational operators: >,<,>=,<=,++,!=.

(iv) In the assignment no.2, use friend function to overload operators: +,-,/,*

(v) Define a class to represent a bank account. Include the following:

a. Data Members:

b. Account n.

c. Name of the account holder

d. Balance amount

e. Member Functions:

f. To assign initial data

g. To deposit an amount

h. To display the data

(vi) Write main() function to test the above class

a. Create two derived classes from already defined class in Assignment 4 (Current

Account and Saving Account). The Saving Account should have a minimum balance.

b. Create a function to withdraw an amount from Saving account. The balance should not

be less than minimum balance after withdrawal.

c. Create a function to withdraw an amount from Current Account. The account allows

the overdraft facility.

(vii) Create a class MAT of size m*n. Define following matrix operations for MAT type objects: Sum, product, transpose

(viii) Define a class String. Use overloaded == operator to compare two string type objects.National Institute of Electronics and Information Technology
'A' Level Course in Information Technology under DOEACC Scheme -Revision 5.1

Write a program that can be used as a database of student's information for a department. The program should be able to dynamically allocate or deallocate storage for the student's records using singly linked list. The database should have the following fields: the first and last names, a course code, and a grade for a student. The program should display the following menu:

a. Welcome to the database menu!

b. Press 1 to insert a new record

c. Press 2 to delete a record

d. Press 3 to search the database (by last name)

e. Press 4 to print a range in the database

f. Press 9 to quit

(ix) Write a program to traverse a singly linked list

a. Merge two sorted linked lists to form a third sorted linked list.

(x) Given a linked list of integers sorted from smallest (at the head end) to largest, make appropriate function that inserts the node in the linked list so that it remains in sorted order.

(xi) Implement a data structure using array that supports the following operations: insert, findMin, findMax, deleteMin, deleteMax, isEmpty, makeEmpty.

a. Note: You must use the following algorithm: maintain a sorted array. Insert new items into the correct position in the array, sliding elements over one position to the right, as needed. findMin and findMax, and deleteMax are trivial. For deleteMin remove the item in position 0, and slide over all the other items one position to the left.

b. Consider a database of patient's information for a hospital. The program should be able

to allocate and deallocate storage memory for the patient's records. The database should have the following information field: the first and last names, patient id, address , related disease , date of admission Devise an appropriate C++ class Queue using arrays to implement the following functions
a) inserting an element(a record) in the queue,

b) deleting the record from the queue, and

c) searching record from the queue by patient id.

(xii) Consider a database of patient's information for a hospital. The database should have the following information field: the first and last names, patient id, address, related disease, date of admission Devise an appropriate C++ class Queue using array to implement the following functions

a) inserting an element (a record) in the queue,

b) deleting the record from the queue, and

c) searching record from the queue by patient id.

(xiii) Consider a database of patient's information for a hospital. The program should be able to allocate and deallocate storage memory for the patient's records. The database should have the following information field: the first and last names, patient id, address, related disease, date of admission Devise an appropriate C++ class Queue using linked list to implement the following functions

a) inserting an element(a record) in the queue,

b) deleting the record from the queue, and

c) searching record from the queue by patient id.

(xiv) Implement stack as class in C++ having a constructor. Create the functions with the functionalities for the operations: push(), pop(), isempty(), isfull().

Display a Main Menu as below:
"Stack Operations Main Menu"

1.Push
2.Pop
3.IsEmpty
4.IsFull
0.Exit

Implement the stack using array.

(xv) Implement stack as class in C++ having a constructor. Create the functions with the functionalities for the operations: push(), pop(), isempty(), isfull().

Display a Main Menu as below:
"Stack Operations Main Menu"

1.Push
2.Pop
3.IsEmpty
4.IsFull
0.Exit

Implement the stack using linked list.

(xvi) Infix, Postfix and Prefix notations are three different but equivalent ways of writing expressions.

Infix notation: X + Y Operators are written in-between their operands. This is the usual way we write expressions. An expression such as A * (B + C) / D is usually taken to mean something like: "First add B and C together, then multiply the result by A, then divide by D to give the final answer."

Postfix notation (also known as "Reverse Polish notation"): X Y + Operators are written after their operands. The infix expression given above is equivalent to A B C + * D /

Prefix notation (also known as "Polish notation"): + X Y Operators are written before their operands. The expressions given above are equivalent to / * A + B C D

Write a program in C++ to convert an infix expression into a) prefix and b) postfix expression?

(xvii) Write a C++ program for the evaluation of postfix expression (created in assignment no 17) using Stack?

(xviii) Build a Binary Search tree by adding the following items into an empty Binary Search Tree, in order: J, T, E, G, F, B, A, R, O, P, U, M, N, K, L, H, D, S, C, Q, I

Write the functions for the tree traversal in each of the following orders:

a) Pre-order traversal

b) In-order traversal

c) Post-order traversal

(xix) Consider a database of student's information for a department. The data should be stored in array of Structures. The database should have the following fields: the first and last names, a course code, and a grade for a student. Create a C++ function to search a particular student's information from the database using linear search

(xx) Given a two-dimensional array, containing names of all officers in an organization in the alphabetical order. Write a C++ program to input the name of an officer and then search the name in the given array. In case the name is found, print the position of the same in the array, else print the message "Search Unsuccessful"". Use binary search technique.

(xxi) Two arrays of size 20 each containing numbers in sorted order. Merge the two lists (arrays) in a fashion that the resulting list (array) also contains the numbers in sorted order.

(xxii) There are different sorting techniques used for arranging the values present in the list. Suppose we have a list of names of persons as a) "Rajan", "Rohit", "Aman", "Jinny", "Sanjay", "Bhatachariya" Write C++ programs arrange these names alphabetically using: a) Bubble Sort, b) Selection Sort, c) Insertion Sort, d) Quicksort

(xxiii) A bank needs to maintain records of its customers. It is decided to create a database using BTree (2-3 tree). The order is based on the Aadhaar number of each Customer. Each record contains the following information. Name, AadhaarNumber. A DBMS needs to be designed to provide menu driven facility to its users. The facilities are:

I. Insert the record for a new customer

F. Find and display the record for a customer specified by name or by Aadhaar number

D. Delete the record of a customer from the database.

(xxiv) Depth First Search is any search algorithm that considers outgoing edges (children) of a vertex before any of the vertex's siblings that is, outgoing edges of the vertex's predecessor in the search. Extremes are searched first. This is typically implemented with a stack. Write a program in C++ to find the DFS of a given graph.

(xxv) Breadth First Search is any search algorithm that considers neighbours of a vertex, that is, outgoing edges of the vertex's predecessor in the search, before any outgoing edges of vertex. Extremes are searched last. This is typically implemented with a queue. Write a program in C++ to find the BFS of a given graph.

A6-R5.1(Computer Organization and Operating Sysytem)

(i) Do as directed below.

a. Convert to decimal: 0x3d7e

b. Convert to binary: 0x3d7e, 0x2ED1

c. Convert from decimal to binary: 764

d. Convert from decimal to hexadecimal: 764

(ii) Do as directed below.

a. Shift operations (left shift and right shift - logical and arithmetic) on single byte quantities. Each of the answers should be 8 binary digits or 2 hexadecimal digits.

Answer the following questions for x = 0x54

b. What is its binary representation?

c. What is x << 3 in binary representation and in hexadecimal representation?

d. What is x >> 2 (logical) in binary representation and in hexadecimal representation?

e. What is x >> 2 (arithmetic) in binary representation and in hexadecimal representation?

(iii) Do as directed below.

a. Design an AND gate and an EX-OR gate using NAND gates

(iv) Do as directed below.

a. What Is Digital Logic?

b. What Is Combinational Logic?

c. Explain Sequential Logic?

(v) Do as directed below.

a. What Are The Developments Of Computer Design?

b. Explain About Program Counter (pc)?

c. What Are Memory Address Register (mar) & Memory Data Register (mdr)?

(vi) Do as directed below.

a. What is an Interrupt Service Routine?

b. Mention what are different types of interrupts in a microprocessor system? Explain?

(vii) There are three components in a course: Quiz (Q), Assignment (A), and Journal (J). You pass the course (P) only if you pass any two or more components. Draw the truth table and design a minimized combinational circuit to show the concept.

(viii) Prove the following Boolean algebra identities:

a. (A+B).(A+C)=A+B.C

b. A+A.B=A

c. (A+B'+A.B)(A+B')(A'.B)=0

d. A+A'.B=A+B

(ix) Given the Boolean function F= a'.b' + a'.d.c' + b.c.d'
a. List the truth table of the function.

b. Draw the logic diagram using the given Boolean expression.

c. Simplify the Boolean expression using Boolean Algebra Identities.

d. Draw the logic diagram using the simplified Boolean expression.

e. Check whether the truth table of part b and of part c is identical.

(x) Perform the arithmetic operations given below with binary numbers using signed 2's complement representation wherever required. Use eight bits to accommodate each number together with its sign.

a. (+12) + (+56)

b. (-35) + (- 49)

c. (-85) – (+71)

d. (-109) + (-11)

(xi) Do as directed below.

a. Getting Help type the 'date' command to display today's date

b. Learn more about this command by accessing its manual page

c. Start a forward search for the word "file" by typing: /file

You can move forward to the next match by pressing n. Also try moving backwards through the matches by pressing N:

d. Execute the following command to display a summary of all man pages that have the

keyword "password" in the description

e. Explore what is, info and –help for getting help

(xii) Try the following commands and write their output:

a. whoami

b. who

c. uname

d. pwd

e. date

f. cal

g. wc

(xiii) Do as directed below.

a. Bash shell maintains a history of the commands that you type. Try 'history' command.

i. Display last 5 commands from history.

ii. Identify how you can recall the previous command.
iii. What does !34 does?

b. Change the history size.

(xiv) Do as directed below.

a. Find the location of the crontab file.

b. How does locate command work?

c. Try find command

d. Find files that were modified (or created) less than 5 minutes ago

e. find files in the /usr directory that are larger than 2MB in size

f. try following command and write its output

g. whereis passwd

h. Explain the difference between locate and find command.

(xv) Do as directed

a. Check your current working directory

b. Make the root directory your current working directory and verify with

the pwd command

c. Change back to your home directory

d. (hint: cd command can be executed without a path)

e. Using an absolute path, change to the /usr/bin directory

f. What do you understand by Absolute and Relative

g. Using a relative path, change to the /usr/share/doc/bash directory

h. Use a relative path to change to the directory above the current directory.

(xvi) Do as directed below.

a. Listing files and directories

Execute the following command

ls -l /etc/hosts

Your output should be similar to the following:
sysadmin@localhost:~$ ls -l /etc/hosts

-rw-r--r-- 1 root root 150 Jan 22 15:18 /etc/hosts
b. So, what does all of this extra output mean?

-

rw-r—r--

1

Root

Root

150

Jan 22 15:18

List contents of subdirectories along with directories
Display hidden files in the /etc directory.

List all files in the /etc directory that are exactly four characters long
Display all files in the /etc directory that begin with the letters a, b, c or d
Display inode numbers of the files.

(xvii) Do as directed below.

a. Make a copy of the /etc/hosts file and place it in the current directory

b. What will be the output if you use -v switch with this command.

c. Explain the purpose of following commands:

i. mkdir

ii. rmdir

iii. cat
iv. touch

v. rm

d. Explain the difference between touch and cat command.

e. How can you append the contents in an existing file

f. Create a file named 'myfile' in your home directory. Create a directory named 'mydir'.
Move 'myfile' inside 'mydir'.

(xviii) Do as directed below.

a. View a large text file with the help of cat command.

b. Use the more command to display the entire contents of the /etc/passwd file.

c. Use less command for the same purpose.

d. What is the difference between more & less command?

e. Use head command to display the contents of /etc/passwd. By default, how many lines will be displayed in the output

f. Try to use tail command.

g. Write the head command to display the first four lines of the /etc/passwd file

h. Write an alternate command to perform the above operation.

i. (Hint : use -n option)

(xix) Do as directed below.

a. _____account is a special user account that has virtually unlimited access and control over the system.

b. To access the root user account, the su or sudo commands are normally used. Use su command to switch user

c. The sudo command is typically used to execute a single command as the root user by prefixing that command with sudo.

d. Which users can use sudo command? How you can grant sudo access to a user

(xx) Using vi editor, do followings.

a. Create a subdirectory named vi_work in your home

b. Create a new file called my_file. Write an essay that should fill the screen. Practice insert, save, and quit.

c. practice using simple search forwards and backwards.

d. Change all occurrences of the word “XXX” to “YYY”

e. Move first six lines to the bottom of the file

f. Create an abbreviation ho to expand to holiday.

(xxi) Do as directed below.

a. Create a new user named 'student' who is a secondary member of the 'research' group and a primary member of their own private group. Use a comment of 'Linux Student' that will appear as the full name of the user when they do a graphical login.

b. Use the passwd command to set the password for the student user and view the shadow file entry for the student user.

c. Use the 'last' command to see if the student has ever logged in.

d. Use the grep command to view the record for 'student' account in /etc/passwd and /etc/shadow files.

e. Explain the various fields of colon delimited /etc/passwd file and /etc/shadow file

f. View account information for your account using id command. Check the same information for root user also.

g. Try to execute following commands:

i. whoami

ii. groups

(xxii) Do as directed below.

a. Create a directory named 'public'. Create a file 'pub-file' inside this directory.

b. Write command to see the permissions on this directory and file

c. use the chmod command with symbolic notation to add write permission for others on 'public' directory

d. Create another directory named 'private'.

e. Use the chmod command with numeric notation to remove any permission from the group or others on the private directory.

f. Grant all users the same read and write permission of the pub-file using symbolic

notation

g. Change the ownership of the file. Also, change the group ownership of the file.

h. Who can change the ownership of the file?

(xxiii) Do as directed below.

a. Use the redirection symbol > to redirect the output from the normal output of stdout (terminal) to a file

b. Append some text to this file

c. Execute following commands

d. find /etc -name hosts 2> err.txt

e. cat err.txt

f. redirect 'stdout' and 'stderr' into two separate files

g. Write a command to convert contents of a file a.txt to lower case

h. List the contents of etc directory and send the output to the more command

i. Extract all of the usernames from a database called /etc/passwd with the help of cut command

j. Display the output of Step 7 in sorted order.

(xxiv) Do as directed below.

a. Create a file named 'source' containing the text "data" by using redirection

b. View the details and inode information of the source file

c. Use the ln command to create a hard link

d. Whether the hardlink file and the original source file share the same inode?

e. What is the link count of the source file

f. Remove the last link that was created and list the files. Is there any change in link count?

g. Create a symbolic link to the source file and view the details of both files? Do both

files have same inode?

(xxv) Do as directed below.

a. From the terminal, type the following command:

b. ping localhost > /dev/null

c. This command is running in “foreground”. Terminate the foreground process by pressing Ctrl-C

d. Start the same process in the background by adding the ampersand & to the end of the command

e. Use jobs command to see the commands running in current terminal.

f. Bring the first command to the foreground

g. Use kill command to stop the process

New Ideas YT
By Satyam Sahu

A7-R5.1(Databases Technologies)

(i) Create a table called EMP with the following structure:

NAME	TYPE
EMPNO	NUMBER(6)
ENAME	VARCHAR2(20)
JOB	VARCHAR2(10)
MGR	NUMBER(4)
DEPTNO	NUMBER(3)
SAL	NUMBER(7,2)
Allow NULL for all columns except ENAME and JOB	

(ii) Add a column commission to the EMP table. (Commission numeric and null allowed.)

(iii) Modify the column width of the job field of emp table.

(iv) Add columns DoB to the emp table.

(v) Insert 5 records into EMP table:

EMPNO	ENAME	JOB	MGR
7369	SMITH	CLERK	7566
7399	AJAY	SALESMAN	7566
7499	ALLEX	SALESMAN	7698
7521	WENNER	SALESMAN	7698
7566	JONES	MANAGER	7839

(vi) Delete the row whose EMPNO is 7566.

(vii) Create a DBMS table named Student containing stdno, stdname, stream, roll. Indicate the primary key.

(viii) Create another table named Std_record containing the fields stdname, stream from the previous table and adding new fields such as admission

(admission year) and graduation graduation year). Which field is the foreign key in this table? Explain.
(ix) How to install MariaDb in a Linux or Unix environment? Install it step by step and mention
the steps.

(x) Create a simple table using MariaDB named Weather, consisting three records as Month, Day and Temperature. Set it as the current default database.

(xi) Clone this table.

(xii) Change the innodb_page_size for the previously created original table.

(xiii) Suppose there are two sets of candidate keys as:
(supplier_no, part_no) and (supplier_name, part_no).
How many types of relations we can establish between them?

(xiv) If there are three sets of established relation from the previous question such as quantity, supplier_name, supplier_no, then under which normal form this table is existing and why?

(xv) Create a database Games in MongoDb and make a collection with name "Play". Show the newly created database.

(xvi) Insert records SL No, Name, Country, into Play collection. Now show the table consisting of these records.

(xvii) Find the document from Play collection where any game's name starts with "C".

(xviii) Create a JSON object country with attribute countryname with value in string and attribute continent in string also.

(xix) Now access these JSON objects and show them.

(xx) Setup the connection for connecting a remote MongoDB in mongo shell environment.

(xxi) Create a table Area. Insert fields as sl_no, area_name, number_of_people, total_voters, no_of_children. Show the table.

(xxii) Drop only one field from any of the fields.

(xxiii) Update a field by changing its area_name.

(xxiv) Now show the latest updated table.

(xxv) Use Maria DB and the following database tables:

STUDENT				
ROLL_NO	NAME	SUBJECT_ CODE	PHONE	EMAIL
1	Meena	E1	000 000 0000	meena@some.com
2	Teena	E2	000 000 0000	reena@ some.com
3	Reena	E1	000 000 0000	reena@ some.com

SUBJECT	
SUB_CODE	SUBJECT_NAME
E1	B.TECH
E2	B.C.A
E3	COMPUTER SCIENCE

a. Create the above tables for Database College and fill in values
b. Query to find all the students name and roll no. enrolled for Subject – "B.C.A"
c. Query to find all email id for student.
d. Query to Update name "Meena" to "Heena" for Roll_No 1.
e. Query to print all columns from both tables having subject "B.Tech".
f. Query to print all students without email and phone

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